MATH 266Z - THE MATHEMATICS OF BPS STATES

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1. Course Syllabus

1.1. **Overview.** This one-semester course introduces the mathematical structures underlying BPS states in classical and quantum field theory. BPS configurations (named after Bogomolny, Prasad and Sommerfeld) form a special class of field configurations, characterized by the property that they obey first order elliptic PDEs as compared to the usual second order Euler-Lagrange PDEs of the standard action functionals in mathematical physics. Examples include gradient-flow trajectories, holomorphic maps, and solutions to the self-dual Yang–Mills equations, which may be viewed respectively as special solutions to Newton's equation with inverted potential $-V$, the harmonic map equation, and the full Yang–Mills equations.

BPS configurations turn out to have a very rich mathematical structure. On the one hand, they are often easier to understand and characterize explicitly, for instance the Atiyah-Drinfeld-Hitchin-Manin construction of self-dual instantons. On the other hand, these configurations often play a key role in defining powerful mathematical invariants in geometry and topology: for instance pseudo-holomorphic maps are the main ingredient in the construction of Lagrangian Floer homology and its extension to the Fukaya category.

The goal of this course is to develop these ideas systematically and to explore the role BPS objects play in modern geometry, topology, and mathematical physics.

1.2. **Outline.** Tentatively, the course will be divided into four parts, roughly organized by the dimension of the underlying field theory.

Part 1 will consist of working through the basic results of Morse theory as formulated by Witten by using gradient flow trajectories. In terms of field theory, Morse

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theory corresponds to studying BPS configurations (the gradient flow lines) in a one-dimensional field theory.

Part 2 will consist of studying solitons and instantons in the setting of Landau-Ginzburg models. This is closely connected to Picard-Lefschetz theory or "holomorphic" Morse theory. Here we will study the Picard-Lefschetz formula and the Cecotti-Vafa wall-crossing formula. We will also outline how these BPS configurations lead to a construction of the Fukaya-Seidel category associated to a holomorphic Morse function.

Part 3, consisting of the three-dimensional phenomenon will be brief, since the subject is less developed here. We will aim to discuss some of the basic PDE's that govern BPS configurations such as the Fueter equation and its massive deformations, and also connect to the subject of three-dimensional mirror symmetry and symplectic duality.

Finally in part 4 we will discuss BPS states and wall-crossing in four dimensions, with the discussion culminating in the celebrated Kontsevich-Soibelman wall-crossing formula.

1.3. **Prerequisites.** A working knowledge of Lagrangian field theory and differential geometry will be helpful. No prior exposure to quantum field theory is required; relevant concepts will be introduced as needed.

1.4. **Textbook and References.** There is no official textbook. Relevant references will be pointed out as we cover the topics. The lectures will be self-contained.

1.5. **Assignments and Grading.** Grades will be based entirely on homework. Approximately one problem set will be assigned every 3–4 weeks.

1.6. **Contact.** Outside of class you can find me at room 116 at the CMSA building (20 Garden Street, Cambridge). Email: ahsankhan@g.harvard.edu. Office hours will be decided at a later point.